Haoran Ma

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Education

Stanford University Sep. 2024 – Dec. 2028 (expected)
Ph.D. Candidate in Physics; M.S. in Physics en route (expected Jan. 2027) GPA: 3.76/4.00
Ph.D. Minor in Computational and Mathematical Engineering (in progress) Minor GPA: 4.00/4.00
Coursework: Machine Learning, CS 229 (A); Bayesian Statistics, STATS 370 (A); Randomized Algorithms and Probabilistic Analysis, CS 265 (A); Investment Science, MS&E 245A (Credit); Optimization, CME 307 (current).

Tsinghua University, Zhili College Sep. 2020 – Jun. 2024
B.S. in Chemistry, Minor in Physics GPA: 3.81/4.00

Research Experience

Stanford University, Philip Bucksbaum Lab (Stanford PULSE Institute) Stanford, CA
Research Assistant *Jan. 2025 – Present*

- Built and lead an experiment on **strong-field ionization by quantum light**: bright squeezed vacuum (BSV) from SHG-pumped parametric down-conversion in a two-crystal SU(1,1) interferometer, superposed with a coherent 800-nm field and focused into a xenon velocity-map-imaging spectrometer with an MCP–Timepix3 detector.
- Built a **1 kHz single-shot pipeline**: a Newton-ring interferometric readout gives the BSV amplitude and phase from one Fourier coefficient per camera frame, computed within the 1-ms gap between pulses, and is paired with the photoelectrons recorded on the same shot.
- With about one detected electron per 30 shots, estimated the field-correlated yield noise with an **unbiased variance estimator** that removes the finite-sample term, **block-bootstrap** errors over 60-s blocks, deconvolution of 10° rms phase jitter, and an out-of-sample fit for slow drift.
- Found that the rms yield noise falls from about 40% to 10% with the measured phase, while its relative variance differs by more than an order of magnitude between ionization channels, reproduced by focal-volume-averaged TDSE calculations with one free parameter (**first-author Letter, submitted to *PRL***).
- Developed an **autoencoder-based workflow** to screen timing-diagnostic traces at the LCLS X-ray free-electron laser (CXI beamline).

Stanford University, Aaron Lindenberg Lab Stanford, CA
Research Assistant *Sep. 2024 – Dec. 2024*

- Contributed experimental data acquisition to a study of non-equilibrium entropy production in a non-Markovian quantum dot (*Nature Physics*, 2026); assisted with setup and characterization for a light-induced liquid-crystal phase-transition project.

Tsinghua University, Jie Yang Lab Beijing, China
Undergraduate Research Assistant *Aug. 2021 – Jun. 2024*

- Studied $Fe(CO)_5$ photodissociation with **MeV ultrafast electron diffraction**: independently built a Python pipeline for **10,000+ noisy diffraction images** and used genetic-algorithm fitting to extract structural dynamics from low signal-to-noise data (**first-author *Nature Communications***).
- Proposed and led SLAC MeV-UED beamtime proposal U103 on retro-Diels–Alder reaction dynamics; helped build a liquid flat-jet sample-delivery system in vacuum.

ETH Zürich, Hans Jakob Wörner Lab Zürich, Switzerland
Summer Intern *Jun. 2023 – Sep. 2023*

- Built a chopper-based timing and data-acquisition workflow for three-channel time-resolved photoelectron spectroscopy, and Python analysis tools for high-throughput and single-shot measurements used during live experiments.

SLAC National Accelerator Laboratory, UED Team Menlo Park, CA
Summer Intern *Jul. 2022 – Aug. 2022*

- **Primary developer of Python acquisition and visualization software** used in SLAC MeV-UED gas-phase user experiments since 2022; implemented low-latency processing to support decisions during live experiments.
- Helped develop a delivery system for high-boiling-point solids in high vacuum.

Independent Projects

Sequence Modeling and Backtest Validation on Equity Data Research notes: Jul. 2026

CRSP US equities 2000–2025 and China A-shares (Tushare); Python, PyTorch, DuckDB

- Built a rolling ridge-regression factor pipeline (**46 million security-day observations**) and trained a two-layer GRU on 120-day sequences, with annual walk-forward evaluation over 2018–2025.
- Traced an apparently strong short-horizon signal to bid-ask bounce, redesigned the target to include spreads, and rejected the model when little signal remained.
- Measured look-ahead bias with as-of joins on announcement dates and survivorship bias against the full CRSP universe with delistings, and tested strategy robustness with temporal splits and 500 randomized controls.

Skills

Experimental: ultrafast Ti:sapphire lasers, nonlinear optics (SHG, parametric down-conversion), squeezed light, interferometry, velocity-map imaging, MCP / Timepix3 detectors, ultrafast electron diffraction, X-ray FEL experiments, high vacuum, timing and synchronization.

Computational: Python (NumPy, pandas, SciPy, PyTorch, Matplotlib), C++, SQL (DuckDB), Mathematica; real-time data pipelines, Fourier-domain signal processing, statistical estimation and bootstrap, TDSE simulation (QPROP), numerical optimization, sequence models, autoencoders.

Publications & Talks

- **H. Ma** et al. "Photoelectrons resolve the phase-dependent variance of bright squeezed vacuum." Submitted to *Physical Review Letters*; arXiv: 2610.31811 (2026). **First author.**
- **H. Ma** et al. "UNWRAP: Single-shot phase analysis of bright squeezed vacuum by Newton-ring interferometry." arXiv: 2610.30005 (2026). **First author.**
- **H. Ma** et al. "Berry pseudorotation enabled photodissociation of $Fe(CO)_5$ observed by ultrafast electron diffraction." *Nature Communications* **17**, 4910 (2026). **First author.**
- Y. Shen, C. Chen, **H. Ma** et al. "Non-equilibrium entropy production and information dissipation in a non-Markovian quantum dot." *Nature Physics* **22**, 374–381 (2026).
- E. Weckwerth et al. (incl. **H. Ma**). "Few-femtosecond time resolution in optically pumped hard X-ray scattering at a free-electron laser." *Optica* **13**(5), 863–866 (2026).
- Co-author of two further preprints (arXiv: 2608.23292, 2605.27612).
- Oral talk, APS DAMOP 2026 (two-color strong-field ionization with non-classical light); poster, FEMTO 15, Berlin (2023).

Honors

Silver Medal, 34th Chinese Chemistry Olympiad (CChO) | Comprehensive Excellence Scholarship, Tsinghua University (2021, 2022, 2023)